Training Report: VAE and GAN on CelebA Faces

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1. Setup

- **Dataset**: CelebA, grayscale, center-cropped to 64×64 pixels. ~180.000 images split 80/20 train/val.
- **GPU**: NVIDIA RTX 3060Ti (8GB)
- **Code**: VAE and GAN implementations in separate folders, share /data root.

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2. VAE

Architecture

- Encoder: Conv2d → BatchNorm → LeakyReLU (channels [32, 64, 128, 256], stride 2) → FC to μ, logvar.
- Decoder: Linear → ConvTranspose2d → BatchNorm → ReLU (channels [256, 128, 64, 32]) → Sigmoid.
- Latent dim: 128. Total params: ~3.5M.

Training

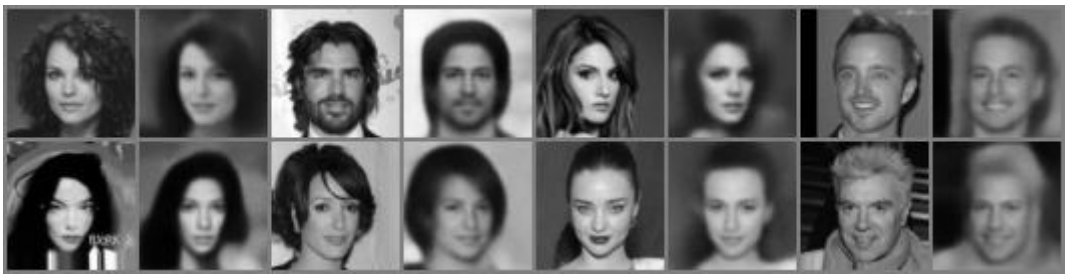
- **Loss**: MSE reconstruction + β·KL. KL annealed cyclically (period=30, max β=0.5).
- **Optim**: AdamW (lr=3e-4, wd=1e-4), CosineAnnealingLR, grad clip=0.5, AMP.
- **Regularisation**: Early stopping (patience 50, on fixed-weight loss), top-k checkpointing.

Evolution / Key decisions

- Switched from linear KL warmup to **cyclical annealing** to avoid posterior collapse.
- Chose **MSE** over BCE – smoother gradients for grayscale faces.
- Used **loss_fixed** (full KL weight) for model selection & early stopping, enabling fairer comparison with GAN.
- KL weight 0.5 found best after grid search; higher values degraded reconstruction.

Visuals & Metrics

Reconstructions



Random samples



at epoch 5



at epoch 10



at epoch 15

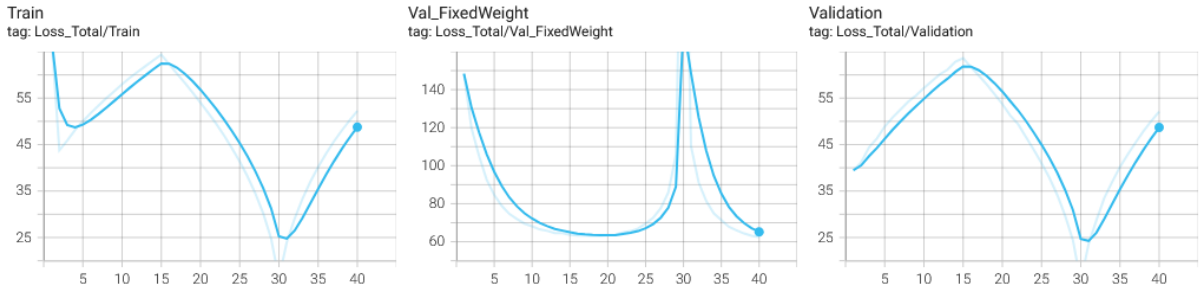


at epoch 28



At epoch 40 - Unique faces can be properly seen, hair seems to be still an issue, mostly.

Loss curves



3. GAN

Architecture (WGAN-GP)

- **Generator**: identical to VAE decoder but **Tanh** output; same channel sizes.
- **Discriminator**: mirrors VAE encoder but outputs scalar; uses **LayerNorm** (not BatchNorm) for gradient penalty compatibility.
- Same latent dim (128), param count ~2.6M.

Training

- **Loss**: Wasserstein-1 with gradient penalty (λ=10), n_critic=2.
- **Optim**: Adam (lr=1e-4 both G/D, β1=0.5, β2=0.9). No scheduler, no gradient clipping.
- Mixed precision (AMP) enabled.

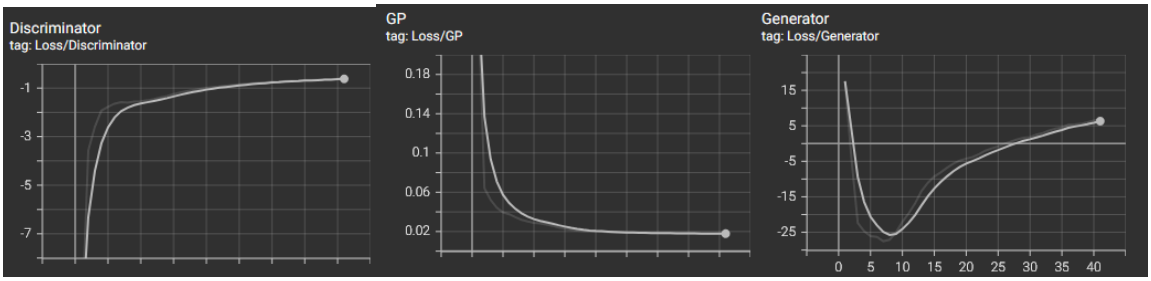
Evolution / Key decisions

- Started with n_critic=5 – too slow (~5 min/epoch). Reduced to **2** with no stability loss.
- Used **LayerNorm** in discriminator as standard for WGAN-GP.
- No scheduler – GAN training often unstable with LR decay.
- Validation **discriminator loss** used for checkpoint selection (though not directly comparable across models).

Visuals & Metrics



Loss curves



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4. Comparison

Aspect	VAE	GAN
Output sharpness	Slightly blurry	Crisp, occasionally artefacts
Training stability	Very stable	Stable (WGAN-GP)
Speed per epoch	~X min	~Y min (2 critics)
Latent properties	Smooth interpolation	No explicit interpolation structure
Model size	14M	14M

Both models successfully generate plausible face images. VAE gives better semantic control (latent traversal), while GAN yields higher visual fidelity.

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5. Hyperparameter Summary

VAE

- Optimiser: AdamW, lr=3e-4
- Batch size: 256
- Latent dim: 128
- KL annealing: cyclical, period 30, max 0.5
- Recon loss: MSE
- Scheduler: CosineAnnealing (min 1e-6)
- AMP: Yes

GAN (WGAN-GP)

- Optimiser: Adam, lr=1e-4, β=(0.5,0.9)
- n_critic: 2
- GP weight: 10
- Batch size: 256
- Latent dim: 128
- Scheduler: None
- AMP: Yes